

Supplement to the Edwards Aquifer J-17 scored test

Amin Abaee

Independent Researcher

ORCID: [0000-0002-0019-1842](https://orcid.org/0000-0002-0019-1842)

Email: amin_abaee@ut.ac.ir

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Supplement to "Does a one-pool water-balance model improve forecasts of

Edwards Aquifer head? A scored test at J-17"

Section S1 carries the independent replication of the uncertainty layer of Section 5.3.1; Section S2 carries the truncated-detail climate fixed-window record of Section 5.4. All values are archived with the analysis code and re-compute from the registered per-origin forecast files.

S1 Independent replication of the uncertainty layer

An independent replication of this layer is registered with the repository: the archived script `campaign_e3_dm_uncertainty.py`, a second Diebold-Mariano + moving-block-bootstrap implementation — HAC lag $h = 1$ with unweighted truncation and population-variance scaling, block length h , 20,000 replications, seed 0, deterministic (byte-identical on re-execution) — reading the same registered per-origin forecast file. Its verified output (archived alongside it) reproduces every load-bearing reading: the M1 retention margin $DM\ z = -0.86$ with block CI $[-1.22, +0.56]$ ft (bootstrap $p = 0.38$); the M2m edge over persistence ($p = 0.001$); the $h = 5$ climatology win ($p = 0.035$). It also registers two rows this table does not carry — the $h = 1$ climatology loss (+2.94 ft against persistence, $p = 0.046$) and the M2 $h = 5$ loss (+12.4 ft, $p < 0.001$) — both consistent with the readings above. The two implementations use different seeds, block lengths, HAC weighting, and variance conventions, and agree on every conclusion: the 0.39-ft AR(1) retention margin is statistically indistinguishable from zero, the M2m edge is the only one-year margin that separates from noise, and the five-year climatology win survives.

S2 Climate-informed modules on the remaining fixed windows

The remaining fixed windows complete the same record: drawdown 1951–56, M2_Renso 24.30, M2_Rprecip 28.67, M2_combo 24.70, precipitation oracle 18.16 ft; pre-permit wet 1991–95, M2_Rprecip 22.03, M2_Renso 23.03, M2_Rar 23.94, M2_combo 25.71, oracle 10.98 ft; critical-period era 2015–23, M2_Rprecip 14.52, M2_Rar 14.67, M2_Renso 16.01, M2_combo 16.57, oracle 9.75 ft — on this one window the two climate-informed modules M2_Rprecip and M2_Rar edge past M1 by about one foot (14.52 and 14.67 versus 15.62 ft). On the recharge target itself the fixed-window scores are an order of magnitude coarser on every window (climate modules 199–937; precipitation oracle 80.7–487, against the 556×10^3 acre-ft climatology scale of the rolling record), so the marginal head advantage is a window-specific result, not a recharge forecast, and it does not

enter the abstract.